

Practical 3

Filter RetailPulse Sales by Region and Date

Related topic: Focus Reports Using Filters | With Analytics, Dashboard and Graphs

Name: _____ Roll No.: _____ Date: _____

Data note: all figures, charts and the dashboard in this report are built from an **illustrative sample dataset** of 1,800 sales rows (Jul-Dec 2026) generated to demonstrate the analysis. Replace them with the numbers from your own RetailPulse report in Cognos before submitting. The sample data is also supplied as a CSV file.

Aim

To scope the RetailPulse Sales list report to a single region and reporting period using a simple filter and an advanced detail filter, to generalize it with a prompt, and to analyse and visualise the filtered result.

Tools Used

IBM Cognos Analytics (Reporting) and the grouped RetailPulse list report built in Practical 2. Analysis and charts were prepared from the exported report data.

Procedure

Step 1: Open the report. Open *RetailPulse Sales* in edit mode, run it unfiltered and note the baseline row count *N*.

Step 2: Add the simple filter. Open the Filters pane and add **Region = "North"**.

Step 3: Add the advanced detail filter. Add a custom filter with scope **Detail** on Sales Date between the quarter start and end dates.

Step 4: Combine with AND. Make sure the connector between both filters is **AND**, so only North, current-quarter rows remain.

Step 5: Run and verify. Confirm every row is North and inside the quarter, and that the row count is smaller than *N*.

Step 6: Convert Region to a prompt. Change the filter to **[Region] = ?Region?** (or tick *Prompt every time the report runs*), save as a prompted report, and run it for North and South.

Filter expressions used

```
-- Advanced detail filter (Q4 2026)
[Sales Date] between {d '2026-10-01'} and {d '2026-12-31'}

-- Combined logic
[Region] = 'North' AND [Sales Date] between {d '2026-10-01'} and {d '2026-12-31'}

-- Prompted version
[Region] = ?Region? AND [Sales Date] between {d '2026-10-01'} and {d '2026-12-31'}
```

For a quarter that updates automatically, replace the literal dates with `_add_months(_first_of_month(current_date), -mod(extract(month, current_date) - 1, 3))` as the start date and three months later minus one day as the end date.

Observations

Row counts recorded at each stage of the practical (sample data):

Stage	Filter applied	Row count	% of unfiltered
Unfiltered report	None	1,800	100%
After Region filter	Region = North	537	30%
After both filters	North AND Q4 2026	277	15%
Prompted run (North)	Region = ?Region?	277	15%
Prompted run (South)	Region = ?Region?	247	14%

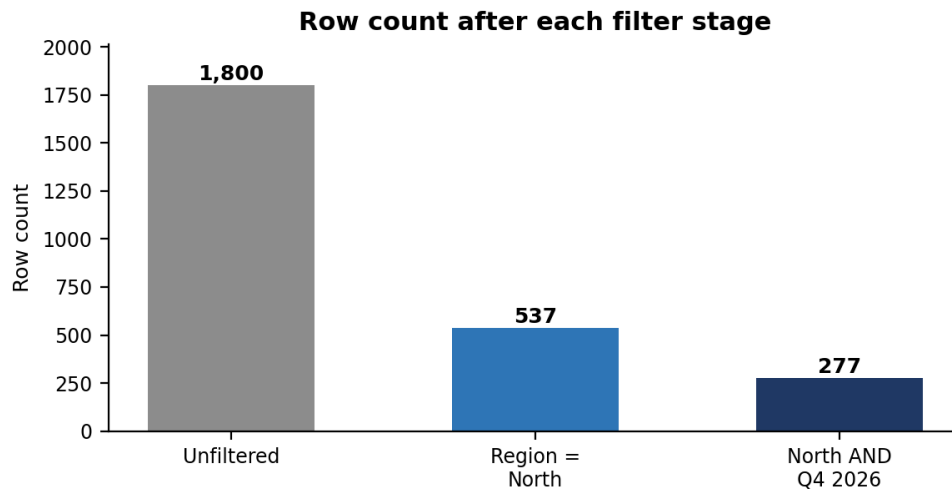


Figure 1: Row count falls from 1,800 to 277 as each filter is applied.

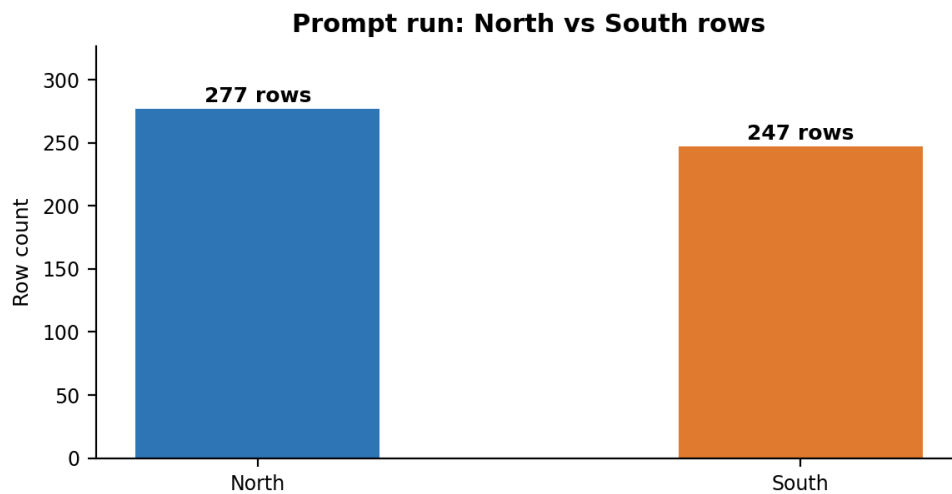


Figure 2: The same prompted report returns different rows for North and South.

Analytics of the Filtered Result

Key figures for the filtered scope (Region = North, Q4 2026):

Metric	Value
Rows in scope	277
Total revenue	305,600
Units sold	1,201
Average revenue per row	1,103
Share of all-region Q4 revenue	29.1%
Row reduction vs unfiltered	84.6%

Q4 2026 comparison across regions

Region	Rows	Revenue	Units	Avg revenue/row	Revenue share
North	277	305,600	1,201	1,103	29.1%
South	247	282,905	1,149	1,145	26.9%
East	192	239,497	888	1,247	22.8%
West	196	223,090	868	1,138	21.2%

Q4 2026 revenue by region

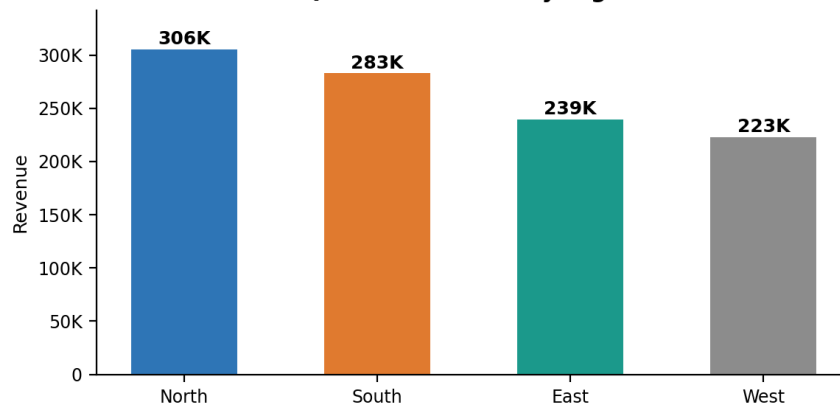


Figure 3: Revenue by region within the target quarter.

North, Q4 2026 by product line

Product line	Rows	Revenue	Units	Share
Electronics	76	152,480	322	49.9%
Home & Kitchen	65	68,413	296	22.4%
Apparel	59	40,784	242	13.3%
Sports	35	32,618	172	10.7%
Grocery	42	11,305	169	3.7%

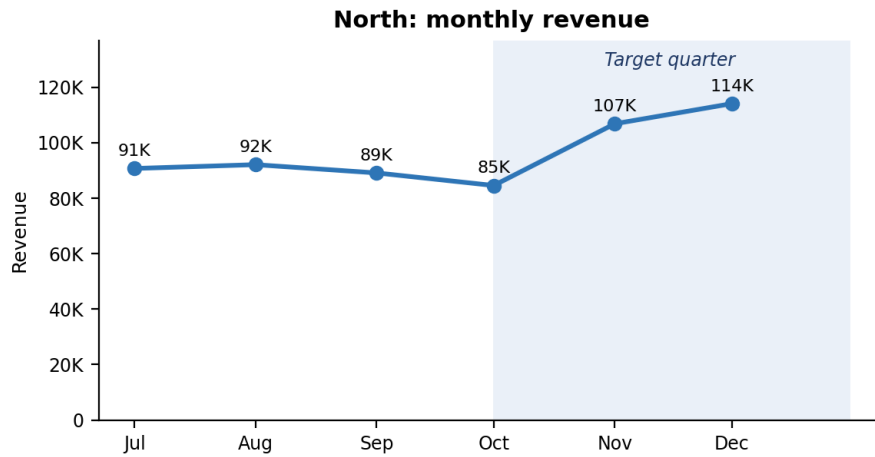


Figure 4: Monthly North revenue; the shaded band is the target quarter.

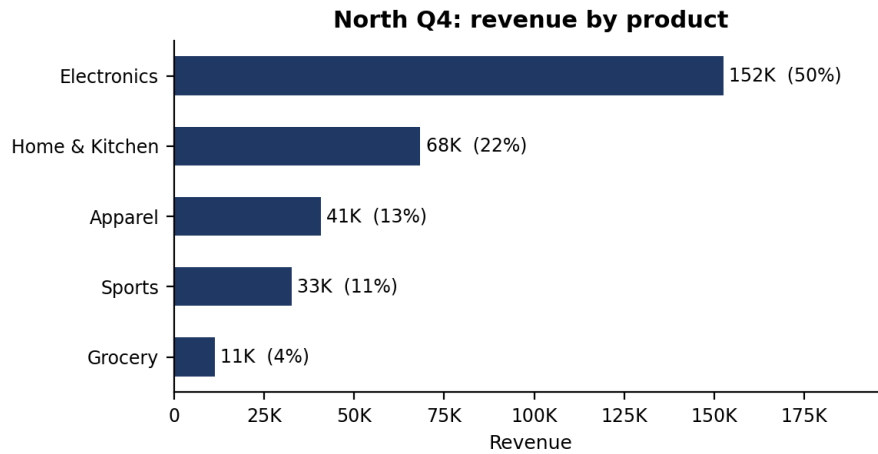


Figure 5: Product-line contribution to North Q4 revenue.

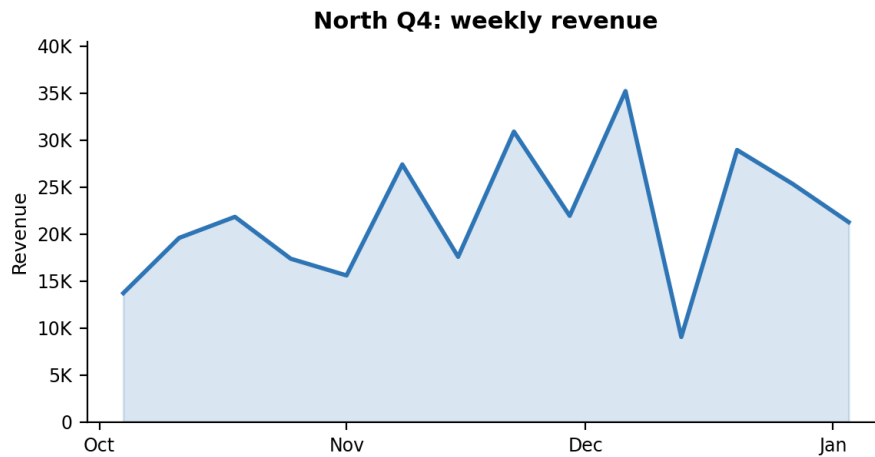


Figure 6: Weekly North revenue across Q4 2026.

Dashboard

RetailPulse Sales Dashboard

Filtered scope: Region = North AND Sales Date in Q4 2026 (1 Oct - 31 Dec) | Illustrative sample data

Rows in scope

277

of 1,800 total

Revenue

306K

29% of all-region Q4

Units sold

1,201

North, Q4

Avg order value

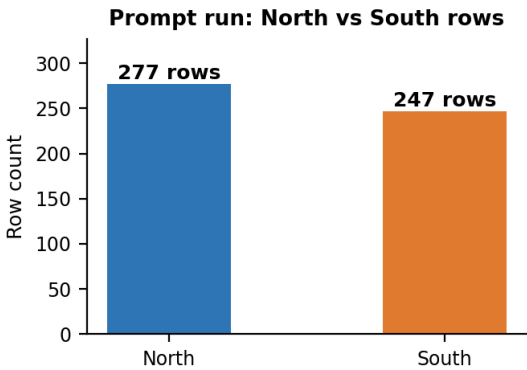
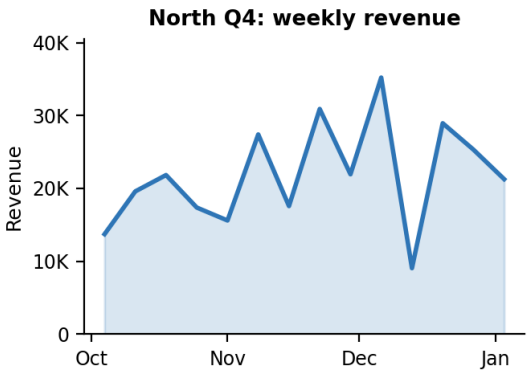
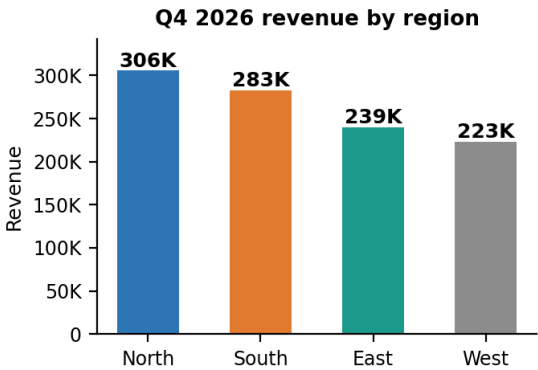
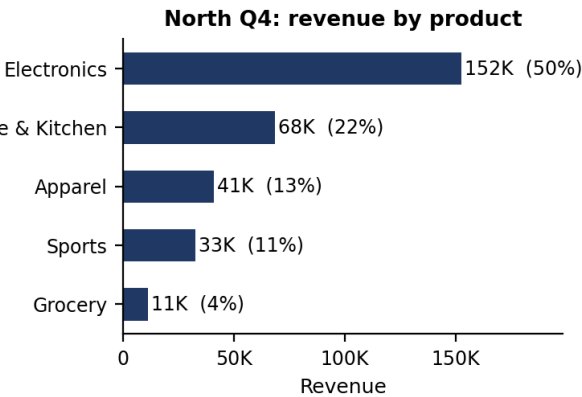
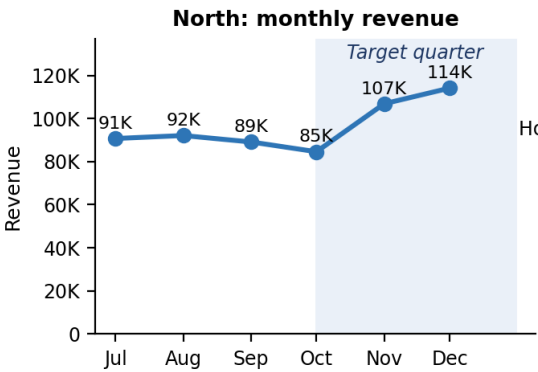
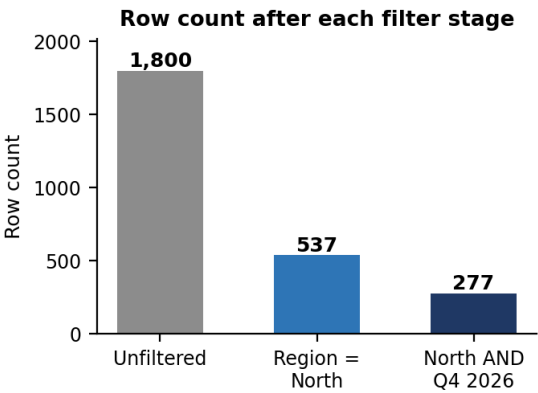
1,103

revenue per row

Row reduction

85%

vs unfiltered report



Key Findings

- The two filters together cut the report from **1,800** rows to **277**, a reduction of 85%. Region alone leaves 537 rows; the date filter then removes the North rows outside Q4.
- North is the largest region in the quarter with **29.1%** of revenue (305,600), followed by South at 26.9%.
- Electronics dominates the North mix at 50% of revenue, so North Q4 performance depends heavily on one product line.
- Monthly North revenue rises from 84,592 in October to 106,871 in November and 114,137 in December, an increase of 35% across the quarter.
- The prompted report returns 277 rows for North and 247 for South with no edits to the report, confirming it is reusable by other managers.

Result

- Only North-region rows within the target quarter remain after filtering.
- The detail filter expression joins the date-range condition with AND.
- The row count is visibly smaller than the unfiltered report.
- After converting Region to a prompt, re-running with South returns South-region data.

Conclusion

A simple filter narrows a report to one value, while an advanced detail filter handles range logic such as quarter dates. Combining them with AND scopes the output precisely, and replacing the fixed Region value with a prompt turns a one-off report into a reusable one. Charting the filtered data shows the scoped result is not just smaller but analytically meaningful: it isolates a region's trend and product mix.

Viva Points

- **Detail vs. summary filter:** a detail filter works on rows before aggregation, while a summary filter works on aggregated results.
- **AND vs. OR:** AND keeps only rows meeting both conditions, while OR returns rows meeting either.
- **Why prompt:** it avoids hard-coding values and lets users choose the region at run time.